

Literature Review:

Sign language serves as a primary communication medium for deaf and hard-of-hearing individuals. Recent advancements in Computer Vision and Natural Language Processing have enabled the development of automated systems capable of translating sign language into text and converting text into sign animations.

Sign language translation: Modern systems use camera-based vision and deep learning to decode signing. Typical pipelines extract hand/pose landmarks (e.g. with OpenPose or MediaPipe) and feed them to CNN/RNN models. Studies report high accuracy on isolated gestures but lower performance on continuous sentences. For example: Zhang & Jiang (2024) note that CNN/RNN methods excel at isolated signs, while continuous sign sequences benefit from Transformer-based models[1], Anturkar et al. (2025) compared 3D-CNN vs LSTM on 1,200 ASL signs: 3D-CNN reached 92.4% accuracy (vs 86.7% for LSTM) but with higher latency[2]. Camgoz et al. (2020) introduced a multi-task Transformer that jointly learns gloss recognition and translation, achieving state-of-the-art results[5]. Continuous sign translation remains hard: Camgoz et al. (2018) report only ~9.6 BLEU score for end-to-end video to text (vs 18.1 BLEU when using a sign-gloss intermediate)[5]

In summary, vision-based deep models (CNN/LSTM/Transformer) and pose tracking are common, yielding up to 90–93% on segmented gestures[1][2], but systems struggle with real-time performance, long sentences, and complex grammar.

Sign language production(text to sign): Earlier work used animated avatars driven by manually defined gestures. These often look robotic and ignored facial cues[3]. For instance, avatar approaches produce sign sequences but lack fluency. Stoll et al. (2020) introduced an end-to-end pipeline that generates continuous sign video from text[4]. Their pipeline: (1) NMT from text to gloss, (2) motion graph to find pose sequence, (3) GAN-based model conditioned on pose to render realistic video. The generated videos achieved very high visual quality: SSIM between synthesized and ground-truth frames was ≈ 0.97 [4], and PSNR/SSIM metrics indicated near-photorealism. This work demonstrated that deep generation can produce multiple signers in HD video without pre-recorded motion capture.

Common pitfalls include missing non-manual cues (eye gaze, facial expression) and grammatical mismatches[4]. Most frameworks use TensorFlow/PyTorch with pose-estimation libraries (OpenPose/MediaPipe) and neural NMT/attention models.

Key findings: AI-driven avatars show promise for automatic translation, but their accuracy is typically reported on limited datasets (often ASL or European sign) and real-time, sentence-level multilingual support (e.g. Nepali Sign Language) remains largely unexplored[4].

Challenges: Sign–text grammar mismatch requires specialized NLP/gloss handling[6]; annotated data and toolkits remain scarce for many sign languages[7]; and generating fully realistic, high-resolution sign video is still difficult for learned systems (native avatars remain smoother).

Research gaps: Most work targets a few (often ASL) languages and isolated vocabularies; generalizing to other sign languages or large-vocabulary contexts is lacking. Few systems offer full real-time “camera to text” and “text to sign” translation with low latency. Finally, capturing non-manual cues (facial expressions, eye gaze) and other multi-channel signals in a unified system is still largely unresolved.

References:

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